Hypersphere-Based Remote Sensing Cross-Modal Text–Image Retrieval via Curriculum Learning

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***Abstract*— Remote sensing cross-modal text–image retrieval (RSCTIR) is a flexible and human-centered approach to retrieving rich information from different modalities, which has attracted plenty of attention in recent years. It remains challenging because the current methods usually ignore the varying difficulty levels of different sample pairs, stemming from the large image distribution difference and the high text similarity in the remote sensing (RS) field. Therefore, in this article, we propose an innovative hypersphere-based visual semantic alignment (HVSA) network via curriculum learning (CL). Specifically, we first design an adaptive alignment strategy based on CL, which aligns RS image–text pairs from easy to hard. Sample pairs with different levels of difficulty are treated unequally, and we obtain a better embedding representation when projecting the features onto the unit hypersphere. Then, to measure the robustness of cross-modal feature alignment on the unit hypersphere, we introduce the feature uniformity strategy. It reduces the occurrence of mismatching cases and improves generalization performance. Finally, we design the key-entity attention (KEA) mechanism to alleviate the problem of information imbalance among different modalities. KEA has the ability to extract information about the key entity which is aligned with textual information. Despite its conciseness, our framework achieves state-of-the-art performance on classical datasets of RSCTIR tasks while enjoying faster inference. The summed recall of HVSA on the RISCD and RSITMD is 120.97 and 198.94, 2.50 and 10.49 points ahead of the current best methods, respectively. Extensive experiments demonstrate the competitiveness of our method. The code has been released at https://github.com/ZhangWeihang99/HVSA.**

***Index Terms*— Adaptive alignment strategy, curriculum learning (CL), feature uniformity, hypersphere, key-entity attention (KEA), remote sensing cross-modal text–image retrieval (RSCTIR).**

I. Introduction

In RECENT years, advances in remote sensing (RS) Earth observation technology have greatly expanded human perception. For instance, the Sentinel satellite accumulates an excess of 10 TB of data each day. The exponential growth of the RS data has improved the prospects for many applications, such as agriculture and vegetation identification, urban management, and natural disaster detection [1], [2], [3], [4]. Due to the limitation of data processing technology, it is still a challenge to search and retrieve useful RS data accurately and efficiently [5], [6]. Automatic RS image retrieval is receiving more and more attention from researchers [7], [8], [9], [10].

Remote sensing cross-modal text–image retrieval (RSCTIR) aims to use text descriptions as queries to retrieve RS images. RSCTIR has the ability to recognize the properties of objects and the relationship between each object, which understands RS imagery at the semantic level [6]. An efficient and accurate RSCTIR method can retrieve flexible, subjective, and specific information in real-time from the massive and constantly updated RS data [11], [12].

The previous mainstream approaches of RSCTIR are the two-stage approaches based on caption matching [13]. The caption is generated for the query image, and the similarity between the generated text and the query text is calculated to achieve cross-modal image–text retrieval. The essence of such methods is text-to-text retrieval, and its retrieval performance is affected by both the quality of image caption generation and the accuracy of text similarity measurement. As opposed to two-stage methods, embedding-based single-stage methods are proposed by Abdullah et al. [14]. The single-stage methods directly measure the similarity by projecting the data of different modalities to the unit hypersphere. The unit hypersphere has been widely used as it contributes to enhancing the stability of training and separability of feature vectors [15], [16], [17]. Therefore, many single-stage methods use the unit hypersphere as the common subspace [13], [18]. Such end-to-end retrieval methods directly align the query text to the RS imagery, which have become the current primary methods for RSCTIR.

Recently, curriculum learning (CL) has shown its powerful capabilities in boosting the generalization performance of various models [19]. The formalization of the CL strategy is proposed by Bengio et al. [20], aiming to mimic the human learning process by learning the elementary concepts earlier and the advanced concepts later. The traditional CL strategy is gradually increasing the complexity of the training data during the training process. It usually implements an easy-to-hard training strategy with the difficulty metric.

Although many advances have been made in the field of RSCTIR, some challenges are still remaining.

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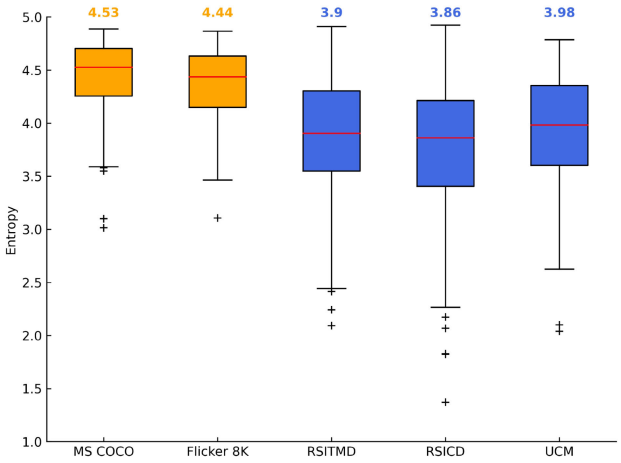


Fig. 1. Box plot of entropy distribution for images from different datasets. To more intuitively demonstrate the differences between batches of data when training the network, we randomly select 128 samples from each dataset. The orange color corresponds to the natural scene dataset, while the blue color corresponds to the RS scene dataset. The red line segments indicate the mean value, which is specifically labeled at the top of each box. The plus sign represents the outlier. The histogram is calculated by dividing the grayscale values into 32 bins. The image entropy values are calculated by OpenCV, and statistics are obtained from Pandas.

First, most previous methods of RSCTIR learn from the natural communities but ignore the differences in data distribution between the RS field and the natural field. There are large image distribution differences and high text similarity in RS scene, which are evidently different from natural scene. As shown in Fig. 1, we use the entropy [21] to perform statistics on a batch of images, including MS COCO [22], Flicker 8K [23], RSITMD [18], RSICD [24], and UCM [25]. For a batch of data, the entropy distribution of RS images has a larger variance and smaller mean. At the same time, Yuan et al. [18] quantitatively analyze the text distribution in RSCTIR using BLEU and METEOR indicators, which can be concluded that the text similarity in the field of RS is significantly greater than that in the field of the nature. The characteristics of wide variation in RS image distribution and the high similarity of representation text distribution expand the heterogeneous gap [26] between multimodal data in RS scene. This challenge leads to an increase in the difficulty of RS image–text features' representation and alignment. The direct adoption of the traditional alignment strategy not only ignores the information that the samples differ in difficulty but also tends to confuse the network.

To address the issue mentioned above, we propose an adaptive alignment strategy based on CL techniques. As shown in Fig. 2, we design a novel learning strategy from easy to hard by considering the degree of difficulty of different sample pairs. This idea of starting easy and adaptively increases the difficulty of learning data stems from CL and leads to better performance.

Second, the current embedded-based single-stage methods mainly focus on the alignment of cross-modal features, ignoring the robustness of alignment [27]. More specifically, the uniformity of features with different semantic information is ignored when projecting multimodal data onto the unit hypersphere. We propose an innovative uniformity loss to reflect the robustness of the multimodal feature alignment. We use the radial basis function kernel to measure the uniformity of different semantic information features, which in turn improves the generalization performance.

Finally, some current methods ignore the problem of information imbalance among multimodal data. Exceeding the information contained in text description queries, RS images contain a tremendous amount of information due to multiscale representation, the presence of small targets, high levels of background information, etc. To alleviate the problem of information imbalance, we design the key-entity attention (KEA) mechanism to extract interesting and significant entity information and filter the redundant information. KEA enables better visual representation for RS image–text alignment.

The contributions of our work can be summarized as follows.

1) Considering the characteristics of data distribution and the varying difficulty levels of different sample pairs in RSCTIR, we introduce an adaptive alignment strategy based on CL. It aligns the RS image–text pairs from easy to hard, which provides useful insights into the RSCTIR task.
2) Beyond the cross-modal feature alignment, we propose the loss of uniformity to measure the robustness of the feature alignment on the unit hypersphere. It reduces the occurrence of mismatching cases and improves generalization performance.
3) The KEA mechanism is devised to address the issue of imbalances between RS image and text information. Visualization experiments have been conducted to demonstrate the effectiveness of the proposed KEA mechanism.

The key objective of our research is to optimize the common unit hypersphere in terms of alignment and robustness by exploiting the previously introduced distributional properties of RS image–text data. Adaptive alignment strategy based on CL and feature uniformity strategy contribute to the effective and robust representation of cross-modal features. Valuable entities are extracted by the KEA, which mitigates the inherent information imbalance. Our proposed hypersphere-based visual semantic alignment (HVSA) delivers outstanding performance while enjoying faster inference. We sincerely hope that HVSA can give some insights to other researchers about RSCTIR.

The subsequent work in this article is organized as follows: in Section II, we briefly introduce the work related to RSCTIR and CL. Then, in Section III, the details of our proposed method and the principles are presented. After that, in Section IV, we conduct extensive comparative and ablation experiments to validate the effectiveness of our method. Finally, in Section V, we conclude our work.

II. Related Work

In this section, we will briefly introduce the development of RSCTIR and CL.

A. RS Cross-Modal Text–Image Retrieval

RSCTIR aims to recall the corresponding RS images using semantic information in the description text. The main

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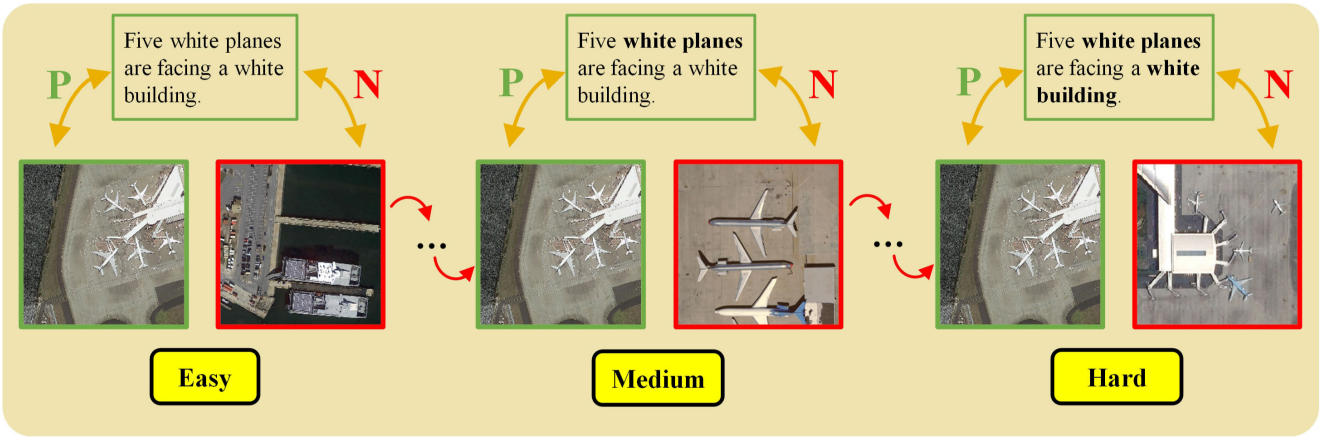


Fig. 2. From left to right, the adaptive alignment strategy based on CL is devised during the training process, taking into account triplet samples at different levels of difficulty. P denotes positive image–text pairs, indicated in green. And N denotes negative image–text pairs, indicated in red. The duplicated information of the unpaired image and the anchor text is marked in bold. The more duplicated information between negative image–text pairs suggests that it is more likely to be mismatched, which implies that the triplet sample is more difficult. During the training process, the adaptive alignment strategy first learns easy triplet samples and then gradually progresses to difficult triplet samples.

challenges of RSCTIR are the heterogeneous gap and RS data distribution's diversity. Different modal representations are inconsistent and located in different feature spaces, so it is challenging to measure the similarity. In addition, large image distribution differences and high text similarity make RSCTIR more difficult. The two-stage approaches based on the caption and the single-stage approaches based on the embedding are the two mainstream approaches to implementing RSCTIR.

The caption-based two-stage methods generate the caption text first. After caption generation, the similarity between the generated text and the query text will be calculated. Qu et al. [25] proposed a deep multimodal network to perform semantic understanding of high-resolution RS images. Lu et al. [24] introduced some principles for a comprehensive description of RS images and contributed a large dataset for RS caption tasks. A sound-active attention network [28] is proposed to generate more targeted subtitles based on the interests of the observer. Zhang et al. [29] fused global and local visual features and filtered out redundant features to obtain more salient features. Wang et al. [30] used multilabeled scene categories to provide sufficient and accurate semantic information. Although the caption-based two-stage RSCTIR methods are relatively well developed, it introduces additional training costs and error amplification.

The embedding-based single-stage methods aim to directly measure the similarity by mapping the data of different modalities into a common subspace. Abdullah et al. [14] proposed a novel RSCTIR framework based on a deep bidirectional triplet network. Yuan et al. [18] tried to explore fine-grained multiscale methods for text–image retrieval and proposed an asymmetric multiscale feature matching network. They also designed a concise and effective cross-modal retrieval model for multiscale and target-redundant RS data [31]. In addition, considering the local information of represents reflects the relationship and salience of the object, Yuan et al. [13] proposed a dynamic fusion module to fuse global and local visual features. Rahhal et al. [32] introduced multilingual text features in addition to English and designed a multilingual framework based on the transformer. Some researchers have also tried to construct the relationship between RS images and text through graph neural networks [27], [33]. Although embedding-based single-stage methods have made some progress, the previous methods mainly focus on improving the alignment of cross-modal features, ignoring the characteristics of RS data distribution and the uniformity of feature distribution in high-dimensional common space. The impact of the varying difficulty levels of different sample pairs is also neglected.

B. Curriculum Learning

Inspired by the human learning process, CL advocates restricting the model to start learning from simple samples and gradually progress to complex ones [19]. The CL strategy can boost the generalization ability of various models, which has been demonstrated in various domains such as object detection, segmentation, and intent detection [34], [35], [36], [37]. Frameworks of CL can be divided into two main categories, data level and model level.

The data-level approaches, which gradually increase the complexity of the training data during the training process, are the current dominant approaches. The traditional data-level CL strategies mainly exploit information about the different difficulties of examples or tasks [38]. To our knowledge, the CL paradigm was first proposed and formalized by Bengio et al. [20]. The predicted likelihood is used by Lee and Grauman [39] to measure the difficulty of the sample in image classification and then learn from the easy samples. Sakaridis et al. [35] gradually increased the difficulty of the semantic segmentation task by attempting to adapt the daytime model to the nighttime images.

The model-level methods gradually extend the capabilities of the model during the training process. Morerio et al. [40] proposed a scheduling strategy for dropout to softly increase the number of units in the network. Sinha et al. [41] convolved the output of each layer of the convolutional neural network (CNN) feature map with a Gaussian kernel whose variance

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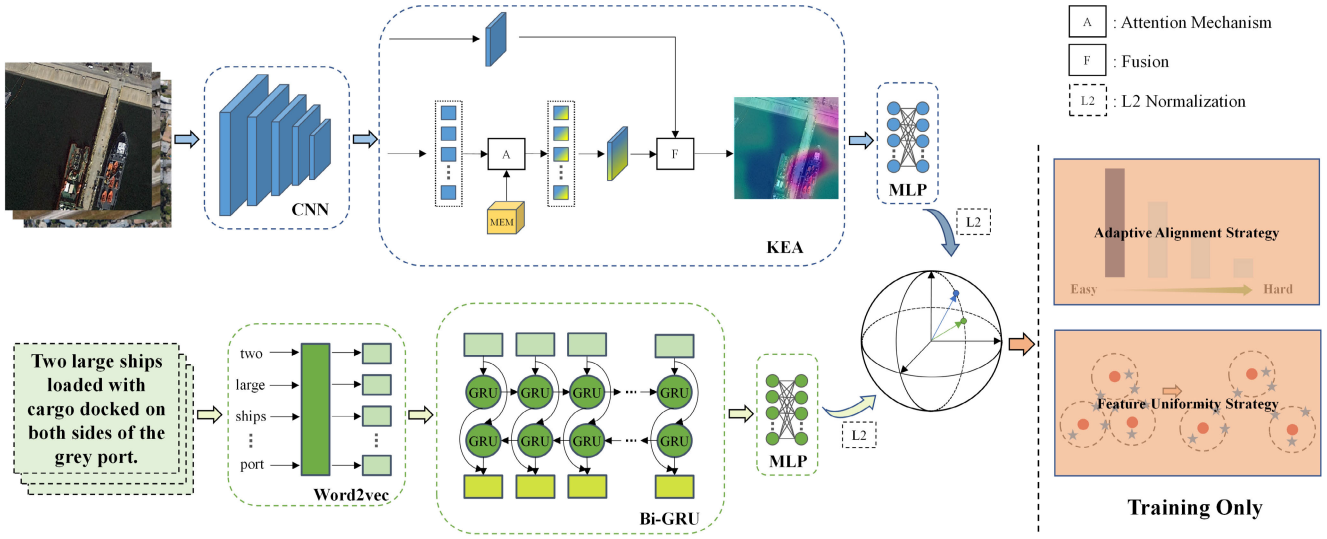


Fig. 3. Proposed HVSA network. In the visual encoding portion, we first use traditional CNNs to extract primitive visual features. Then, based on the primitive features, the KEA is applied to capture the valuable visual features. In the text encoding portion, a classical recurrent neural network is used for semantic representation. For visual-semantic alignment, an adaptive alignment strategy based on CL is applied so that the model follows the principle of learning from simple to complex data. Furthermore, the feature uniformity strategy is enforced to ensure that features with different semantics are uniformly distributed on the hypersphere.

decreases gradually during training. Some anticurriculum methods are also proposed such as hard example mining [42], [43], showing improved results in certain conditions. The CL strategy has shown its powerful capabilities in boosting the generalization performance of various models [19].

III. Method

We show the overall structure of HVSA in Fig. 3. For clarity of description, we mainly introduce our method from four aspects: formulation in Section III-A, KEA mechanism in Section III-C, and the properties of the unit hypersphere in Section III-B. Finally, we introduce the optimizing strategy in Section III-D.

A. Formulation

For RSCTIR, features of input image I and features of input sentence S are obtained after network feature extraction. The obtained sentence features and image features are projected to the unit hypersphere for similarity measurement, which can be represented as

$$i = f\left(I; \theta_f\right) \in \mathbb{R}^{D_c} \tag{1}$$

$$s = g\left(S; \theta_g\right) \in \mathbb{R}^{D_c} \tag{2}$$

$$\mathrm{Sim}(I, S) = i \odot s \tag{3}$$

where $f(\cdot)$ and $g(\cdot)$ represent extraction and projection of the image and sentence, respectively. θ_g and θ_f are the model parameters. i and s represent the image features and sentence features, respectively, projected to the unit hypersphere whose dimension is D_c. $\mathrm{Sim}(I, S)$ is the cross-modal similarity which is calculated by the inner product of features i and s.

1) Visual Representation: We use the traditional CNNs for the initial extraction of RS visual features. We perform transfer learning using pretrained weights. The global visual features obtained

$$i^g = \mathrm{CNN}(I). \tag{4}$$

The query text describes only a portion of the area in the RS image, which means the amount of information contained in images is much greater than the text. To alleviate the problem of information imbalance, we design the KEA mechanism to suppress the distracting information in the image

$$i^k = \mathrm{KEA}\left(i^g\right) \tag{5}$$

where i^g represents the global feature extracted by CNN and i^k is the key-entity feature obtained by KEA. Details of KEA are introduced in Section III-C. Before projecting to the common unit hypersphere, we fuse the key-entity feature with the global feature

$$i = \mathrm{Proj}\left(i^g, i^k\right). \tag{6}$$

2) Text Representation: For a sentence S containing m words, each word corresponds to a real-valued vector [44] via an embedding matrix

$$e_n = W_e(w_n), \quad n \in [1, m] \tag{7}$$

where w_n denotes the nth word. W_e denotes the word embedding layer, and e_n denotes the embedding vector of the nth word.

Considering the contextual information among the sentences, we use a bidirectional gated recurrent unit (GRU) [45] to extract semantic features, with each word input to the bidirectional GRU in order at different time steps. The semantic features of the whole sentence are the average of the hidden features of each word. Then, the semantic features of the whole sentence are projected to the common unit hypersphere. The above process is formulated as follows:

$$h_n^f = \mathrm{GRU}^f\left(e_n, h_{n-1}^f\right) \tag{8}$$

$$h_n^b = \mathrm{GRU}^b\left(e_n, h_{n+1}^b\right) \tag{9}$$

$$s = \mathrm{Proj}\left(\frac{1}{m}\sum_{n=1}^{m}\frac{h_n^f + h_n^b}{2}\right) \tag{10}$$

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where h_n^f and h_n^b indicate the forward hidden features and backward hidden features at step n, respectively. GRU^f and GRU^b denote the forward and the backward GRUs, separately.

B. The Unit Hypersphere

1) Adaptive Alignment Strategy Based on CL: The alignment loss function is defined directly by the similarity between positive and negative samples. When the input is a triplet sample, the positive pairs are supposed to be as close as possible, and the negative pairs are supposed to be as far away as possible. What is more, the distance between negative pairs should be greater than positive pairs with a fixed margin.

The sum of the hinge (SH) loss is the current dominant loss function for RSCTIR

$$T_{\hat{I}_n} = \left[\alpha - \text{sim}(I, S) + \text{sim}(\hat{I}_n, S)\right]_+ \tag{11}$$

$$T_{\hat{S}_n} = \left[\alpha - \text{sim}(I, S) + \text{sim}(I, \hat{S}_n)\right]_+ \tag{12}$$

$$L_{\text{sh}} = \sum_{\hat{I}_n} T_{\hat{I}_n} + \sum_{\hat{S}_n} T_{\hat{S}_n} \tag{13}$$

where S is the query sentence, and I is the paired image. $\hat{I}_n$ is the nth unpaired image. α serves as a margin parameter. $\text{sim}(I, S)$ is the similarity of one positive pair, and $\text{sim}(\hat{I}_n, S)$ is the similarity of one negative pair. $[x]_+ \equiv \max(x, 0)$. The SH loss comprises two symmetric terms. The first item is to iterate over all unpaired images $\hat{I}_n$, given query sentence S. The second item is to iterate over all unpaired sentences $\hat{S}_n$, given query image I. If the similarity between I and S is greater than any negative pairs on the unit hypersphere with a margin of α, the sum hinge loss will be zero.

Inspired by hard example mining, some researchers have also proposed max hinge (MH) loss [43] based on the SH loss, specified in terms of the hardest negatives

$$L_{\text{mh}} = \max\left(T_{\hat{I}_n}\right) + \max\left(T_{\hat{S}_n}\right). \tag{14}$$

Similar to SH loss, MH loss has two symmetric terms. Different from SH loss, MH loss only considers the hardest negative pairs, which means the winner takes all the gradients.

When the SH loss is adopted, the process of gradient backpropagation can be expressed as

$$\begin{aligned}\frac{\partial L_{\text{sh}}}{\partial \theta} &= \frac{\partial\left(\sum_{\hat{I}_n} T_{\hat{I}_n} + \sum_{\hat{S}_n} T_{\hat{S}_n}\right)}{\partial \theta} \\ &= \sum_{\hat{I}_n \neq \hat{I}_m} 1 \times \frac{\partial T_{\hat{I}_n}}{\partial \theta} + \sum_{\hat{S}_n \neq \hat{S}_m} 1 \times \frac{\partial T_{\hat{S}_n}}{\partial \theta} \\ &\quad + 1 \times \frac{\partial T_{\hat{I}_m}}{\partial \theta} + 1 \times \frac{\partial T_{\hat{S}_m}}{\partial \theta}\end{aligned} \tag{15}$$

where θ is the parameter that needs to update the gradient. $T_{\hat{I}_n}$ and $T_{\hat{S}_n}$ represent the difficulty of the triplet sample. The larger the value of $T_{\hat{I}_n}$ represents the more difficult triplet data, which means that the similarity of the positive pairs (S, I) is smaller and the similarity of the negative pairs $(S, \hat{I}_n)$ is larger. For the query sentence S, $\hat{I}_m = \arg\max \text{sim}(\hat{I}_n, S)$, which is the hardest unpaired image. The SH loss considers all the sample pairs equally. The contribution of all the sample pairs to the gradient update is the same regardless of the value of $T_{\hat{I}_n}$ and $T_{\hat{S}_n}$.

When the MH loss is adopted, the process of gradient backpropagation can be expressed as

$$\frac{\partial L_{\text{mh}}}{\partial \theta} = 1 \times \frac{\partial T_{\hat{I}_m}}{\partial \theta} + 1 \times \frac{\partial T_{\hat{S}_m}}{\partial \theta}. \tag{16}$$

Same as the SH loss, $\hat{I}_m$ and $\hat{S}_m$ represent the hardest negative for the positive pair data (S, I). The MH loss only considers the hardest pairs. Only the most difficult sample pairs contribute to the gradient updates of the model parameter θ.

Nevertheless, as for RSCTIR, the wide variation in RS image distribution and the high similarity of representation text distribution lead to an increased difficulty of the alignment. Therefore, directly adopting the traditional alignment strategy may confuse the network, which is not conducive to the generalization performance. To learn more efficient parameters and obtain better generalization performance, sample pairs of different difficulty should be used unequally inspired by CL techniques.

More specifically, our core idea is that sample pairs of different difficulty have different weights, and the network parameters θ are supposed to be learned by the easy-to-hard strategy. Therefore, in gradient backpropagation, we give larger weights to simpler sample pairs, aiming to learn simpler sample pairs first. And the difficulty of the triplet sample is measured by $T_{\hat{I}_n}$ and $T_{\hat{S}_n}$. The process of gradient backpropagation can be expressed as

$$\begin{aligned}\frac{\partial L_{\text{dwh}}}{\partial \theta} &= \sum_{\hat{I}_n \neq \hat{I}_m} \left(1 - \beta T_{\hat{I}_n}\right) e^{-\beta T_{\hat{I}_n}} \times \frac{\partial T_{\hat{I}_n}}{\partial \theta} \\ &\quad + \sum_{\hat{S}_n \neq \hat{S}_m} \left(1 - \beta T_{\hat{S}_n}\right) e^{-\beta T_{\hat{S}_n}} \times \frac{\partial T_{\hat{S}_n}}{\partial \theta} \\ &\quad + \left(1 - \beta T_{\hat{I}_m}\right) e^{-\beta T_{\hat{I}_m}} \times \frac{\partial T_{\hat{I}_m}}{\partial \theta} \\ &\quad + \left(1 - \beta T_{\hat{S}_m}\right) e^{-\beta T_{\hat{S}_m}} \times \frac{\partial T_{\hat{S}_m}}{\partial \theta}\end{aligned} \tag{17}$$

where hyperparameter β represents the emphasis of the model on simple samples.

As Fig. 4 shows, simpler sample pairs have larger weights and have a larger contribution to the gradient backpropagation. Apparently, the difficulty-weighted hinge (DWH) loss is proposed

$$L_{\text{dwh}} = \sum_{\hat{I}_n} T_{\hat{I}_n} e^{-\beta T_{\hat{I}_n}} + \sum_{\hat{S}_n} T_{\hat{S}_n} e^{-\beta T_{\hat{S}_n}}. \tag{18}$$

The proposed DWH loss is capable of adaptively adjusting the weights of different samples in a kind of easy-to-hard principle during the training process. When $\beta = 0$, the proposed DWH loss degenerates to the SH loss, which means that the SH loss is a special case of our proposed DWH loss.

2) Feature Uniformity: Many previous outstanding works focus more on the improvement of cross-modal features' alignment and ignore another important property of cross-modal features which is uniformity.

In our opinion, to obtain robust alignment results, feature vectors with different semantic information should be distributed as uniformly as possible on the hypersphere. When

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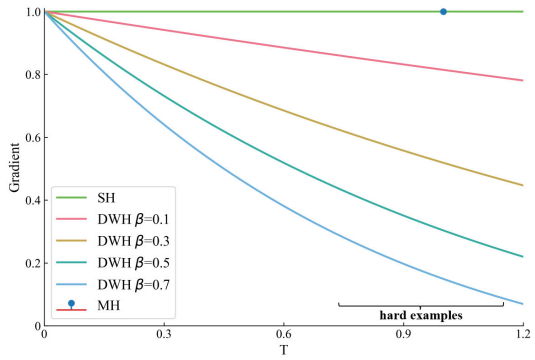


Fig. 4. Gradient contribution of SH loss, MH loss, and DWH loss. The horizontal axis represents the triplet loss which also reflects the level of difficulty of samples during model training. The vertical axis indicates the magnitude of parameter gradients during the update process. α is taken to be 0.2.

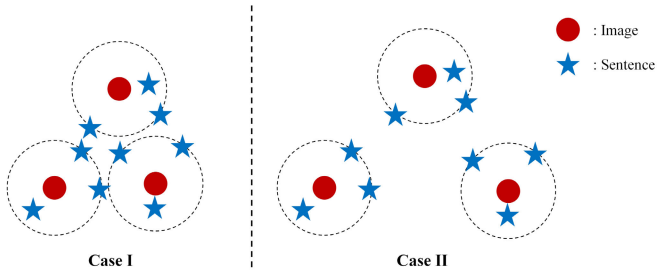


Fig. 5. Two toy examples. We project image and text features from the unit hypersphere onto a 2-D plane. Assuming that all pairs of image and text samples have been correctly matched, two different cases are shown separately.

the cross-modal features are correctly aligned on the unit hypersphere, the alignment loss function value is zero. However, previous outstanding works do not further measure the robustness of alignment. As shown in Fig. 5, the second case is superior to the first, yet both the cases lack any contribution to parameter updating.

We consider the use of the radial basis function kernel to measure the distance between different features on the unit hypersphere. This is because the average pairwise radial basis function kernel is nicely combined with the uniform distribution on the unit hypersphere [17]

$$G(i_u, i_v) = e^{-\frac{\|i_u - i_v\|_2^2}{2\sigma^2}} \tag{19}$$

where i_u, i_v represent different image feature vectors, and σ is a hyperparameter.

Aiming to measure the robustness of the alignment, we propose the uniformity loss function (UNIF). We define the uniformity loss as

$$L_{\text{unif}} = \sum_{u=1} \sum_{v=u+1} e^{-\gamma \|i_u - i_v\|_2^2} \tag{20}$$

where hyperparameter γ represents the degree of penalty for feature distribution uniformity. L_{unif} restricts the image features containing distinct semantic information to moving away from each other on the unit hypersphere, ultimately leading to a more robust alignment of cross-modal features.

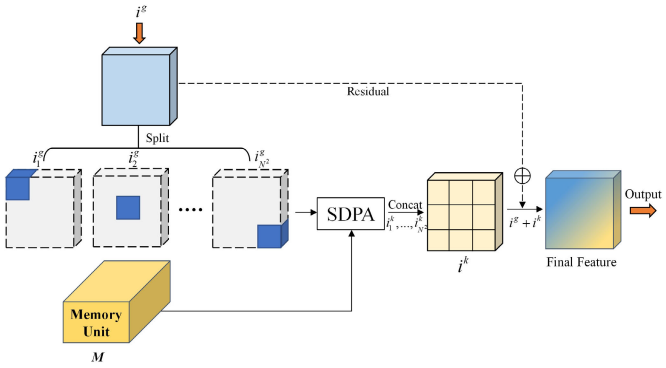


Fig. 6. Proposed KEA module. We first divide the input features into grids to obtain features for each image patch. And then, we use a memory unit to store key-entity information in the image, which helps extract essential information and filter out redundant information through scale dot-product attention (SDPA) [46]. We finally concatenate the image patch features containing key entity information and residually connect them with input features.

The adaptive alignment strategy focuses on matching cross-modal features with the same semantic information. The feature uniformity strategy is targeted at optimizing the distribution of image features. In particular, under the adaptive alignment strategy, the image and text embedding vectors with similar semantic information are gradually closer to each other. Under the feature uniformity strategy, the distance of embedding vectors with distinct semantic information gradually expands. Finally, the text embedding vectors and image embedding vectors are not only correctly aligned but also uniformly distributed on the hypersphere. Notably, the feature uniformity strategy alone cannot determine the similarity between embedding vectors, so it needs to be combined with the adaptive alignment strategy so that the embedding vectors with different semantic information can be distributed as uniformly as possible.

C. KEA Mechanism

Some previous methods directly project the image features to the common unit hypersphere, which is suboptimal for RS images. On one hand, multiscale RS images have lots of variable background disturbance information. The foreground of one scene may be the background of another scene which results in reduced retrieval accuracy. And the large variance of RS image distribution makes it difficult to learn well feature representation. On the other hand, the amount of information contained in the sentence is a subset of the image. The salient targets in the image, which may have no semantic association with the sentence, are not conducive to the cross-modal semantic alignment. Therefore, for the above problems, we design the KEA module to capture valuable information in the images, as shown in Fig. 6.

Specifically, to get the features of key entity in RS images, we first divide the input RS image into an $N \times N$ grid to get N^2 image patches [47], and then obtain the global features i^g through the feature extraction network. Since the distribution of RS images varies greatly, we hope to store the features of typical regions in the whole dataset through memory module M. And then we introduce M as a memory

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module [48] for the key-entity information of the whole dataset. It can measure the importance of different image patches. The feature of the image patches is used as the query vector to calculate the similarity with the typical region in the memory module M to get the feature of the key entity. The memory module M has independent parameters that can be learned while training. The process mentioned above can be expressed as

$$i^g = \left[i_1^g, i_2^g, \ldots, i_{N^2}^g\right] \tag{21}$$

$$a_t = \text{softmax}\left(i_t^g {M_k}^T\right), \quad t \in [1, N^2] \tag{22}$$

$$i_t^k = a_t M_v \tag{23}$$

where i_t^g means the feature of the tth image patch, and M_k is the key vector of the memory module M. The inner product of query vector i_t^g and memory key vector M_k yields the attention score a_t via the softmax function. The feature of the key entity i_t^k can be calculated by the product of attention score a_t with memory value vector M_v. Finally, we concatenate all the features to get the KEA map i^k

$$i^k = \left[i_1^k, i_2^k, \ldots, i_{N^2}^k\right]. \tag{24}$$

Consider some RS scenes with low entropy and no significant target. It is difficult to extract their key-entity features from the memory module. An example is the bare land, a scene with no significant regional features, where global features dominate. Thus, we use the residual connection [49] to fuse the global feature and KEA map and project to the unit hypersphere

$$i = \text{Proj}\left(i^g \oplus i^k\right) \tag{25}$$

where $\oplus$ represents the elementwise addition.

D. Optimizing Strategy

The final objective function is a weighted sum of the DWH loss and the uniformity loss, which can be represented as

$$L = w_1 L_{\text{dwh}} + w_2 L_{\text{unif}}. \tag{26}$$

The two hyperparameters w_1 and w_2 correspond to the fixed weights assigned to the DWH loss and uniformity loss. They serve to balance the relative contribution of each loss during the training process.

We expect HVSA to focus primarily on cross-modal feature alignment and secondarily on feature uniformity, which is automatically tuned during the training process. Manual adjustment of the weight hyperparameters w_1 and w_2 during training is time-consuming and labor-intensive, and using a fixed weight for different training stages is suboptimal. Therefore, after analyzing the convergence characteristics of different losses, we use multilayer perceptrons as an automatic weighting module (AWM) to learn the weights of different loss functions at different training stages

$$L = w_1(t) L_{\text{dwh}} + w_2(t) L_{\text{unif}} \tag{27}$$

where $w_1(t)$ and $w_2(t)$ represent the weights obtained by AWM which is varying with different training time steps t.

The training procedure of the proposed HVSA is shown in Algorithm 1. In the training phase, RS image-text pairs are required. For the image embedding representation, we first extract RS image features using pretrained ResNet-18 [49] as the backbone, divide the image features into $N \times N$ grids, and extract the key-entity features by KEA. The sentence embedding representation is extracted directly by bidirectional GRU. The triplet loss is calculated to measure the difficulty of the triplet sample. The adaptive alignment strategy aligns the RS image–text pairs from easy to hard. The feature uniformity strategy restricts the features to be distributed as uniformly as possible on the unit hypersphere. The two strategies are weighted by AWM to optimize the weight hyperparameters.

Algorithm 1 HVSA Training Procedure

Require:

RS training dataset $D = \{(I_1, S_1), (I_2, S_2), \ldots\}$ (I_n, S_n) is the positive paired data,

Image Encoder $i = f\left(I; \theta_f\right) \in \mathbb{R}^{D_c}$:

Pretrained CNN $i^g = \text{CNN}(I)$,

KEA module $i^k = \text{KEA}(i^g)$,

Visual Projection Layer $i = \text{Proj}\left(i^g, i^k\right)$.

Sentence Encoder $s = g\left(S; \theta_g\right) \in \mathbb{R}^{D_c}$:

GRU module $h = \text{GRU}(S)$,

Text Projection Layer $s = \text{Proj}(h)$,

Automatic Weighting Module $l = \text{AWM}(l_1, l_2; \theta_m)$,

Repeat until convergence:

for each batch $I, S \in D$ **do**
  // RS Image Feature Extraction
  $i^g = \text{CNN}(I)$
  $i^k = \text{KEA}(i^g)$
  $i = \text{Proj}\left(i^g, i^k\right)$
  // Text Feature Extraction
  $h = \text{GRU}(S)$
  $s = \text{Proj}(h)$
  // Adaptive Alignment Strategy
  $l_1 = L_{dwh}(i, s)$
  // Feature Uniformity
  $l_2 = L_{unif}(i)$
  // Auto Search
  $l = \text{AWM}(l_1, l_2; \theta_m)$
  Update θ_f, θ_g, θ_m, **by** l
end for
return i, s, l_1, l_2, l

IV. Experiments' Results and Analysis

In this section, the dataset and evaluation metrics will be introduced first. After that, we introduce some implementation details. We then compare and analyze different RSCTIR methods to validate the competitiveness of our proposed model. Furthermore, a lot of ablation experiments are conducted to systematically analyze the impact of each component in the proposed HVSE. We perform visualization experiments and analysis. Finally, time consumption is compared.

A. Dataset and Protocols

As shown in Fig. 7, RSICD [24] and RSITMD [18] are two common datasets for RSCTIR. Each image sample in the

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1. A bridge is on a yellow ripples.
2. The wide, long bridge cast a shadow over the muddy river.
3. The wide, long bridge cast a shadow over the muddy river.
4. A bridge is on a yellow ripples.
5. A bridge is on a yellow ripples.

(a)



1. There is a purple plane on the runway.
2. The plane with purple fuselage is on the runway.
3. The purple plane is on the runway.
4. A purple plane is on the runway.
5. There is a purple plane on the runway.

(b)

Fig. 7. Samples from RSICD and RSITMD datasets. Each RS image corresponds to five sentences. (a) Sample from RSICD. (b) Sample from RSITMD.

dataset corresponds to at most five described sentences. There are two other commonly used datasets, UCM Captions and Sydney, which we do not use because their sample sizes are too small, only 2000 and 600, respectively.

The RSICD dataset has 10 921 images, and each image sample size is 224 × 224. The RSITMD dataset has 4743 image samples, each of which is 256 × 256, and the semantic representation is more fine-grained than RSICD. In our experiments, the data are divided into training set, validation set, and test set in the ratio of 8:1:1, which is consistent with [13].

We choose the widely used $R@k$ and rsum [50] as the evaluation metrics for model performance. $R@K$ is the ratio of ground truth to the top K recall samples, and the common values for K are 1, 5, and 10. rsum is the sum of various $R@k$, which more intuitively reflects the overall performance of the model.

B. Implementation Details

All the experiments are implemented in PyTorch [51] and conducted on a single NVIDIA 32G V100 GPU. For the reproducibility and credibility of the experiments, we fixed three random seeds and performed three consecutive experiments to take the average. For cross-modal retrieval, the same method with different data augmentations obtains different performance. Data augmentation is not the main focus of this study. Therefore, for the sake of fair comparison, our method adopts the same type of data augmentations as the previous outstanding methods. The data preprocessing and feature space dimensionality are consistent with GaLR [13]. For data preprocessing, we uniformly scale the images from different datasets to 256 × 256, and our data enhancement methods conclude rotation and flipping. The embedding dimension of the single word vector is 300, the image embedding vector and the text embedding vector are 512 dimensions, and the margin of the adaptive alignment strategy is set to 0.2. The hyperparameters β and γ are set to 0.5 and 2, respectively, obtained by grid search. We choose AdamW [52] as the optimizer. We totally train 50 epochs and set the batch size to 256. The learning rate is warmed up to 0.0005 and decayed at a rate of 0.1 per 20 epochs.

C. Comparisons With Several Recent State-of-the-Arts

The proposed HVSA model is compared with previous advanced RSCTIR models on RSICD and RSITMD, and the results are shown in Table I.

1) VSE++ [43]: VSE++ focuses more on difficult sample pairs. The alignment of cross-modal features is measured by the MH loss function.
2) SCAN [53]: SCAN improves the VSE++ model by proposing a cross-attention mechanism between image regions and words. SCAN tries to correspond words in sentences to regions in images.
3) CAMP [54]: CAMP proposes cross-modal adaptive messaging, which adaptively controls the flow of information for cross-modal messaging.
4) AMFMN [18]: AMFMN uses multiscale image features to guide text features' representation, and the query incorporates keywords in addition to text.
5) GaLR [13]: GaLR proposes a multiscale information dynamic fusion module to integrate global and local visual features, which are extracted by graph neural networks and CNNs.
6) HyperMatch [33]: A hypergraph network is designed for modeling the relationships of complex features extracted by CNN. HyperMatch proposes a dynamic multiscale feature fusion module to integrate visual features which are applied to guide text features.

On the RSITMD dataset, our method achieves the best results compared with the previous methods. The rsum of our method is 198.94, which is an improvement of 10.49 points compared with GaLR-MR. It is important that our framework is concise but achieves better results. For visual features, some methods such as GaLR and Hypermatch use Faster R-CNN [55], [56] and graph neural networks for visual features while we only use ResNet-18. The state-of-the-art performance of our model is due to the optimization for feature uniformity on the unit hypersphere, as well as the fact that we take into account samples with different difficulties and improve the learning strategy. An interesting phenomenon is that our method is not the best in the $R@1$ metric, but we are the best in all other metrics. This further demonstrates the effectiveness of the adaptive alignment strategy based on CL, where we sacrifice a portion of the performance on the most difficult metrics $R@1$ to improve the overall performance of our method. The excellent performance of our method on RSITMD demonstrates the competitiveness of our method for unit hypersphere space optimization and learning strategy optimization.

Our method also achieves the best result on the RSICD dataset, where rsum of our model is 120.97, an improvement of 2.50 points compared with HyperMatch. For visual feature extraction and fusion, we use simple ResNet-18 and KEA mechanisms, respectively. Compared with HyperMatch and GaLR, we do not perform complex cross-modal feature fusion and data postprocessing but still achieve competitive results

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TABLE I
COMPARISONS OF DIFFERENT METHODS; PERFORMANCE ON RSICD AND RSITMD TEST SETS

Approach	RSICD dataset							RSITMD dataset						
	Sentence Retrieval			Image Retrieval			rsum	Sentence Retrieval			Image Retrieval			rsum
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++	3.38	9.51	17.46	2.82	11.32	18.10	62.59	10.38	27.65	39.60	7.79	24.87	38.67	148.96
SCAN t2i	4.39	10.90	17.64	3.91	16.20	26.49	79.53	10.18	28.53	38.49	10.10	28.98	43.53	159.81
SCAN i2t	5.85	12.89	19.84	3.71	16.40	26.73	85.42	11.06	25.88	39.38	9.82	29.38	42.12	157.64
CAMP-triplet	5.12	12.89	21.12	4.15	15.23	27.81	86.32	11.73	26.99	38.05	8.27	27.79	44.34	157.17
CAMP-bce	4.20	10.24	15.45	2.72	12.76	22.89	68.26	9.07	23.01	33.19	5.22	23.32	38.36	132.17
MTFN	5.02	12.52	19.74	4.90	17.17	29.49	88.84	10.40	27.65	36.28	9.96	31.37	45.84	161.50
LW-MCR(b)	4.57	13.71	20.11	4.02	16.47	28.23	87.11	9.07	22.79	38.05	6.11	27.74	49.56	153.32
LW-MCR(d)	3.29	12.52	19.93	4.66	17.51	30.02	87.93	10.18	28.98	39.82	7.79	30.18	49.78	166.73
AMFMN-soft	5.05	14.53	21.57	5.05	19.74	31.04	96.98	11.06	25.88	39.82	9.82	33.94	51.90	172.42
AMFMN-fusion	5.39	15.08	23.40	4.90	18.28	31.44	98.49	11.06	29.20	38.72	9.96	34.03	52.96	175.93
AMFMN-sim	5.21	14.72	21.57	4.08	17.00	30.60	93.18	10.63	24.78	41.81	**11.51**	34.69	54.87	178.29
GaLR	6.50	18.91	29.70	5.11	19.57	31.92	111.71	13.05	30.09	42.70	10.47	36.34	53.35	186.00
GaLR-MR	6.59	19.85	31.04	4.69	19.48	32.13	113.78	**14.82**	31.64	42.48	11.15	36.68	51.68	188.45
HyperMatch	7.14	20.04	31.02	**6.08**	20.37	33.82	118.47	11.73	28.10	38.05	9.16	32.31	46.64	165.99
HVSA	**7.47**	**20.62**	**32.11**	5.51	**21.13**	**34.13**	**120.97**	13.20	**32.08**	**45.58**	11.43	**39.20**	**57.45**	**198.94**

TABLE II
ABLATION EXPERIMENTS ON THE RSITMD TEST SET

Fintune	SH	MH	DWH	UNIF	AWM	KEA	Sentence Retrieval			Image Retrieval			rsum
							R@1	R@5	R@10	R@1	R@5	R@10	
	✓						10.03	26.77	39.97	8.63	32.16	51.36	168.92
		✓					5.68	17.85	28.25	4.45	17.35	28.60	102.18
✓	✓						12.61	29.54	41.87	10.14	37.07	57.39	188.62
✓			✓				12.83	30.83	43.93	10.78	36.65	56.52	191.54
✓			✓	✓			**14.68**	30.75	43.58	10.83	37.54	56.06	193.44
✓			✓	✓	✓		13.79	**32.23**	45.13	10.34	38.36	57.39	197.24
✓			✓	✓	✓	✓	13.20	32.08	**45.58**	**11.43**	**39.20**	**57.45**	**198.94**

which demonstrate the superior generalization ability of our model.

D. Ablation Studies

In this section, we perform a systematic ablation experiment to explore the effect of different components of our method. For the credibility of the experimental results, we choose to conduct ablation experiments on two classical datasets, RSITMD and RSICD. The experimental results are summarized in Tables II and III.

As shown in Table II, when freezing the parameters of the backbone, in sentence-to-image retrieval task using SH loss is 4.18%, 14.81%, and 22.76% higher than using MH loss, respectively, on the $R@1$, $R@5$, and $R@10$ indicators. The use of SH loss is also 4.35%, 8.92%, and 11.72% higher than the use of MH loss, respectively, in the image-to-sentences' task. We consider that the main reason for this discrepancy is that the MH loss based on hard example mining is not applicable to the RSCTIR scenario. For RSCTIR, there are large differences in RS image distributions and high similarity in RS text descriptions. Using MH loss does not improve the generalization performance of the model but increases the difficulty of alignment for RSCTIR. By fine-tuning the backbone network, the overall performance of the model rsum can be improved by 19.70 points. The natural scene distribution differs greatly from the RS scene distribution, and the pretraining parameters in the natural scene have room for improvement in the RS scene.

With the CL strategy, the overall performance rsum is improved by 2.92 points compared with the SH loss model after using DWH loss. We do not consider samples with different levels of difficulty equally, allowing the model to focus on training data from simple to difficult, improving model performance. By introducing the alignment robustness loss function, the overall performance is improved by 4.82 points compared with SH loss only. Our method obtains a more robust representation of cross-modal feature alignment and improves the generalization of the model.

Adaptively changing the weights of the loss function at different time steps can lead to an improvement in the effectiveness of the model. Learning the weight parameters by AWM not only reduces manual consumption but also achieves better results. Compared with SH loss only, our proposed loss function and training strategy achieve 8.62 points improvement without adding other feature extraction or fusion modules. KEA improves the overall performance of the model to 198.94. Better alignment is obtained by KEA to suppress the interference information in the images and extract the regions with significant semantic information. The results of the ablation experiments on the RSICD dataset are shown in Table III, which is similar to the RSITMD dataset and will not be repeated here for simplicity.

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TABLE III
ABLATION EXPERIMENTS ON THE RSICD TEST SET

Finetune	SH	MH	DWH	UNIF	AWM	KEA	Sentence Retrieval			Image Retrieval			rsum
							R@1	R@5	R@10	R@1	R@5	R@10	
	✓						5.28	16.83	26.65	4.46	18.02	31.33	102.57
		✓					2.75	9.27	16.29	1.95	9.52	16.43	56.21
✓	✓						6.03	18.91	30.01	5.16	19.59	33.74	113.44
✓			✓				6.19	19.03	31.35	5.32	20.00	34.05	115.94
✓			✓	✓			7.14	20.43	31.32	5.33	20.53	33.41	118.16
✓			✓	✓	✓		**7.84**	20.28	31.11	5.45	20.49	**34.39**	119.56
✓			✓	✓	✓	✓	7.47	**20.62**	**32.11**	**5.51**	**21.13**	34.13	**120.97**

TABLE IV
GRID SEARCH EXPERIMENTS ON THE RSITMD DATASET

DWH	UNIF	Sentence Retrieval			Image Retrieval			rsum
β	γ	R@1	R@5	R@10	R@1	R@5	R@10	
0.1	1	12.17	30.97	44.99	10.83	38.17	57.37	194.50
0.3	1	12.61	31.05	44.69	10.84	37.33	57.38	193.90
0.5	1	13.05	30.83	44.91	11.27	38.33	56.93	195.32
0.7	1	13.05	31.49	**45.87**	11.34	**38.38**	56.65	196.78
0.1	2	12.61	32.55	45.28	**11.42**	37.57	56.92	196.35
0.3	2	**14.09**	31.56	44.25	11.37	37.32	56.37	194.96
0.5	2	13.79	32.23	45.13	10.34	38.36	**57.39**	**197.24**
0.7	2	12.91	**32.67**	45.80	10.72	37.79	55.59	195.48

Combining the experimental results on the two datasets, there are some phenomena worth noting: First, the MH loss based on difficult case mining does not achieve the expected results in the RSCTIR scenario. We suggest one of the main reasons is that the diverse distribution of RS images and the large similarity of RS text descriptions lead to too many difficult samples, which greatly increases the difficulty of the RSCTIR task, so the model is prone to be confused and difficult to converge to a good performance. In addition, the CL strategy is effective on the RSCTIR task. Based on the CL strategy using different levels of attention for different difficult samples, our method achieves competitive results on the RSCTIR task. Finally, our proposed improved CL-strategy-based alignment loss function and alignment robustness loss function are optimized for the unit hypersphere, which does not increase any model parameters.

E. Grid Search

The difficulty attention coefficient β and feature uniformity penalty coefficient γ in the HVSA affect the final performance. β and γ are two important hyperparameters for aligning cross-modal features on the unit hypersphere. As shown in Figure 4, β represents the emphasis of the model on simple samples, where β being larger indicates that the model focuses more on simpler samples and neglects more difficult ones. γ represents the degree of penalty for feature distribution uniformity. We use grid search for hyperparameter tuning, as shown in Table IV.

The AWM is used to learn the weights for alignment and uniformity loss. Our method achieves the best overall performance when β is set to 0.5 and γ is set to 2, as shown in Table IV.

F. Visualization Experiments

1) Distribution Visualization: To provide a more intuitive reflection of the distribution of multimodal features on the unit hypersphere, we reduce the dimensions of our image and text embedding vectors. Specifically, we use the t-distributed stochastic neighbor embedding (t-SNE) technique [58], a similarity data visualization method that preserves both local structures of the data and reveals important global structures. We select 50 images and their corresponding 250 captions from the test set of the RSITMD dataset, which is kept consistent for all the methods to ensure a fair comparison.

Before alignment, as shown in Fig. 8(a), we can clearly see the heterogeneous gap, which means that it is difficult to align data from different modalities directly and there are clear boundaries. The feature representation obtained by the GaLR method is shown in Fig. 8(b). We can see that most of the image embeddings and text embeddings are fused, but five outliers are difficult to match with any text embeddings. Compared with previous outstanding GaLR, our method has better generalization results. Intuitively, the multimodal feature distribution of our method is more uniform, and the number of outlier image data is less. This further proves the superiority of the adaptive alignment strategy and the improvement of the uniformity of feature distribution in hyperspace.

2) KEA Visualization: To address the problem of imbalanced information between RS image and text data, the KEA module is designed to suppress irrelevant information in the image and extract regions with valuable semantic information. To obtain the features of the key entity in the RS images, we first partition the input RS image into an $N \times N$ grid to obtain N^2 image patches and then perform attention weighting.

As shown in Fig. 9, we randomly selected four RS images from different scenes, each showing three KEA feature maps corresponding to the three queries on the right side of the figure. Fig. 9(a) demonstrates that KEA has the capability to recognize objects such as airports, aircraft, and runways. The information in query 1 is almost fully matched. In queries 2 and 3, KEA more focuses on the terminal building and aircraft, with only a small part of the runway. Fig. 9(b) shows that KEA captures the typical entity features such as the bridge, riverbank, and river. As for query 2, KEA focuses more on vehicles on the bridge and part of the river. The KEA of query 3 covers only one side, although there are trees on both the sides. Fig. 9(c) shows an interesting

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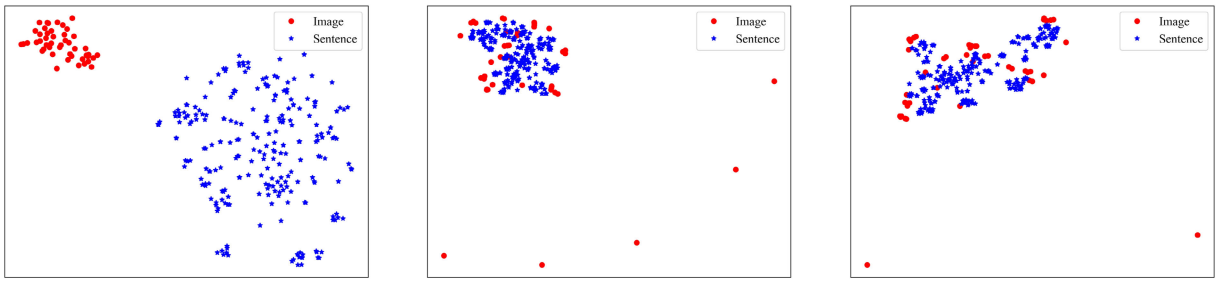


Fig. 8. Feature visualization. Using the t-SNE algorithm, we embed the image and text features onto a 2-D plane from the hypersphere surface. The results are shown in the figure, where (a) represents the unaligned RS image feature and text feature, (b) represents the result of the GaLR, and (c) represents the result of the proposed HVSA.

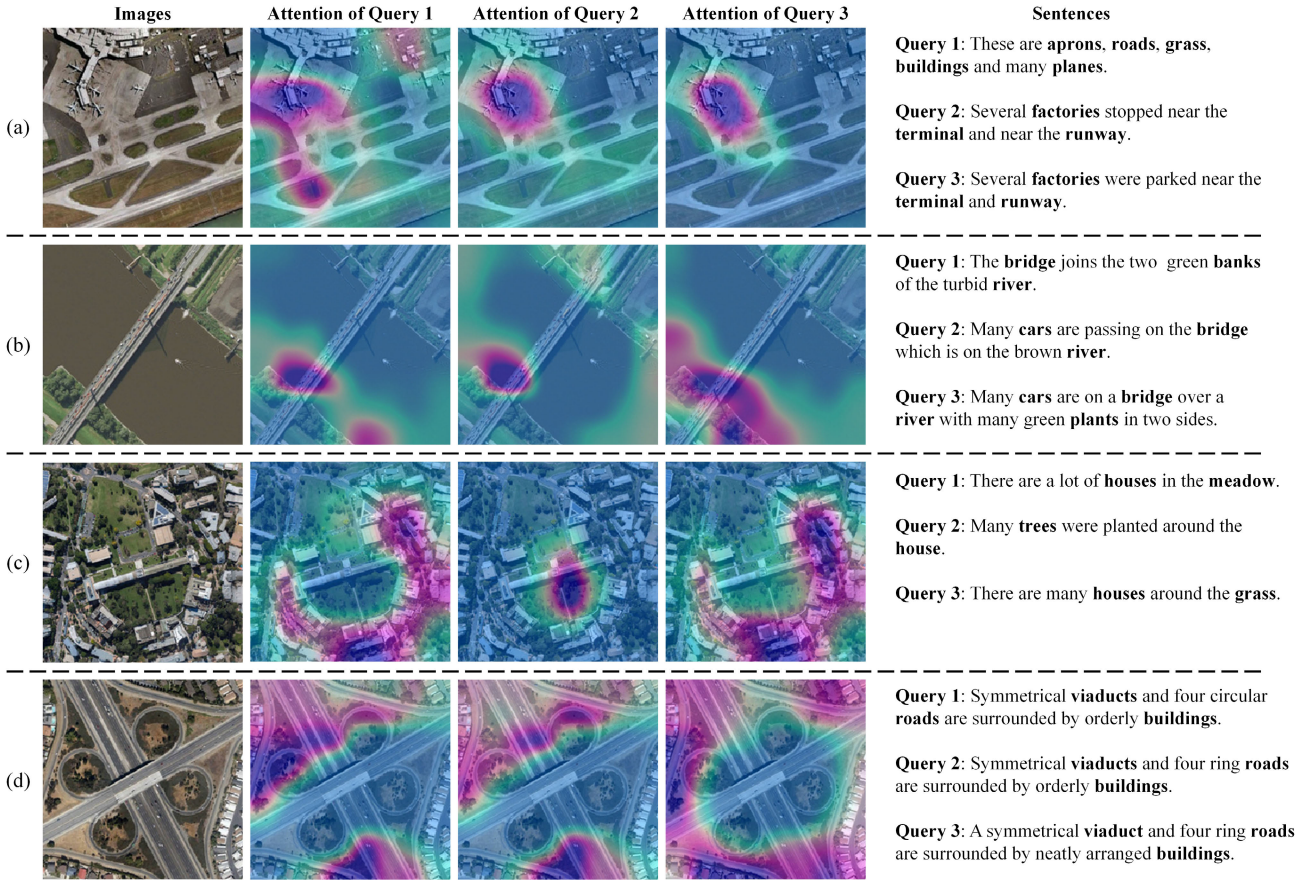
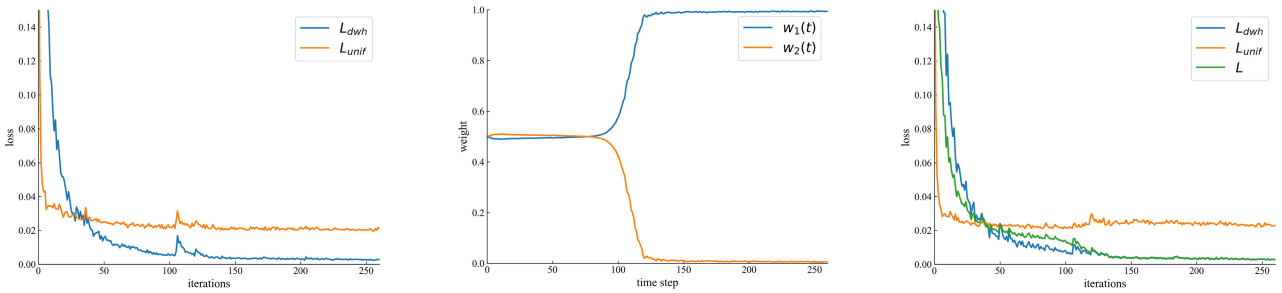


Fig. 9. Visualization of the final visual representation obtained through the KEA mechanism on the RSITMD test set. Each image shows three visual attention heatmaps [57] corresponding to three text queries, respectively. Visual attention heatmaps were obtained by the Grad-CAM technique. KEA captures both the key entities and semantic information in the scene. (a) Visual Attention Heatmaps about the airport. (b) Heatmaps about the bridge. (c) Heatmaps about the residential area. (d) Heatmaps about the viaduct.

phenomenon where queries 1 and 3 mainly describe houses and our KEA captures irregular building areas well. Query 2 focuses more on describing trees, so the key entity features change accordingly. This demonstrates that our KEA can maintain consistency with the text information. The results in Fig. 9(d) show that we adequately capture the valuable information in the queries: overpass and building. However, the text describes four circular roads, while KEA only focuses on one or two, demonstrating that it is not sensitive to quantity.

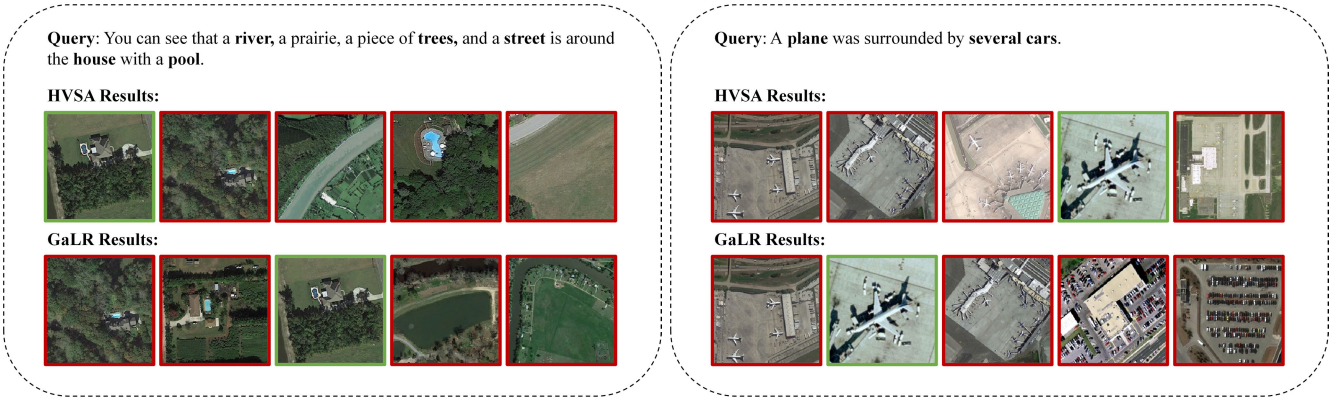
3) Visualization of Losses and Weights Hyperparameters: We devise the AWM to automatically learn the weight hyperparameters at different time steps. Because we expect HVSA to focus primarily on cross-modal feature alignment and secondarily on feature uniformity, and to be able to automatically adjust during the training process. Fig. 10(a) illustrates the variation in loss function values using fixed weight hyperparameters at different time steps. The feature uniformity loss converges to a larger value quickly and the DWH loss converges more slowly to a smaller value. The DWH loss is still in the descent phase when the feature uniformity loss starts to converge, and the DWH loss is lower than the feature uniformity loss. We suggest that it is reasonable to give a larger weight to the DWH loss that has not yet converged when the feature uniformity loss has converged.

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(a) (b) (c)

Fig. 10. Losses and weight hyperparameters' visualization. The trends of losses for fixed weight hyperparameters, weight hyperparameters obtained by AWM at different time steps, and the variation in loss function after automatic weighting are shown, respectively. (a) Losses with fixed weights. (b) Weights at different time steps. (c) Overall loss variation.



(a) (b)

Fig. 11. Results of HVSA and GaLR for text-to-image retrieval task on the RSITMD test set. Key entities in the query text are highlighted. We exhibit the top five retrieved images, indicating correct matches with green boxes and incorrect matches with red boxes. (a) Image to Text Retrieval Case A. (b) Image to Text Retrieval Case B.

Fig. 10(b) illustrates weight hyperparameters obtained by the AWM at different time steps. Before the 100th time step, $w_1(t)$ and $w_2(t)$ vary slowly around 0.5. After the 100th time step, the feature uniformity loss gradually converges, its weight gradually decreases, and the weight of the DWH loss function gradually increases. Fig. 10(c) illustrates the overall loss function variation with automatic weighting. It is noteworthy that the feature uniformity loss is still in convergence despite the small weights obtained by the AWM in the post-training period.

G. Analysis of Retrieval Results

For an intuitive representation of the cross-modal features' alignment on the unit hypersphere, the retrieval result of the proposed HVSA is shown and analyzed. We selected the previously best-performing GaLR model as a comparison with our method. GaLR proposes a dynamic fusion module to fuse global and local visual features, which are extracted by graph neural networks and CNNs. We analyzed these two tasks by selecting several representative examples, as shown in Figs. 11 and 12.

Fig. 11 demonstrates the accurate retrieval performance of HVSA. Fig. 11(a) displays the five most relevant RS images retrieved from the text content. The contained key entities are the river, the prairie, the tree, the street, and the house with a pool. The retrieved results indicate that the proposed HVSA has superior capabilities in text–image retrieval. Compared with GaLR, HVSA not only retrieved the RS images that match the query text at rank 1 but also retrieved other images that contain the key entities described in the text. GaLR's rank 4 result is a pond, which is inconsistent with the textual description, further validating the competitiveness of our method.

Fig. 11(b) demonstrates the robustness of HVSA matching. The query text describes an airplane surrounded by some cars, where the information subject is the airplane rather than the cars. Compared with GaLR, although HVSA correctly matched it at rank 4, all the retrieval results from HVSA contained the information subject airplane, which indicates a highly relevant match with the query text. Moreover, even though GaLR has a correct match on rank 2, GaLR has obvious mismatches, resulting in a lower robustness than HVSA. This further validates the robustness of our method.

Fig. 12 demonstrates the task of retrieving relevant captions based on the RS image. Fig. 12(a) shows retrieval results of HVSA at ranks 1–3 match the query image, and the results at ranks 4 and 5 also contain the key entity "baseball field" of the query image. As shown in Fig. 12(b), three of five retrieval results contain true values. Although the result at rank 2 is a mismatch, it is highly relevant to the query image as well. In contrast, GaLR's rank 2 and rank 3 results contain hillside which is unrelated to the query image. The superior

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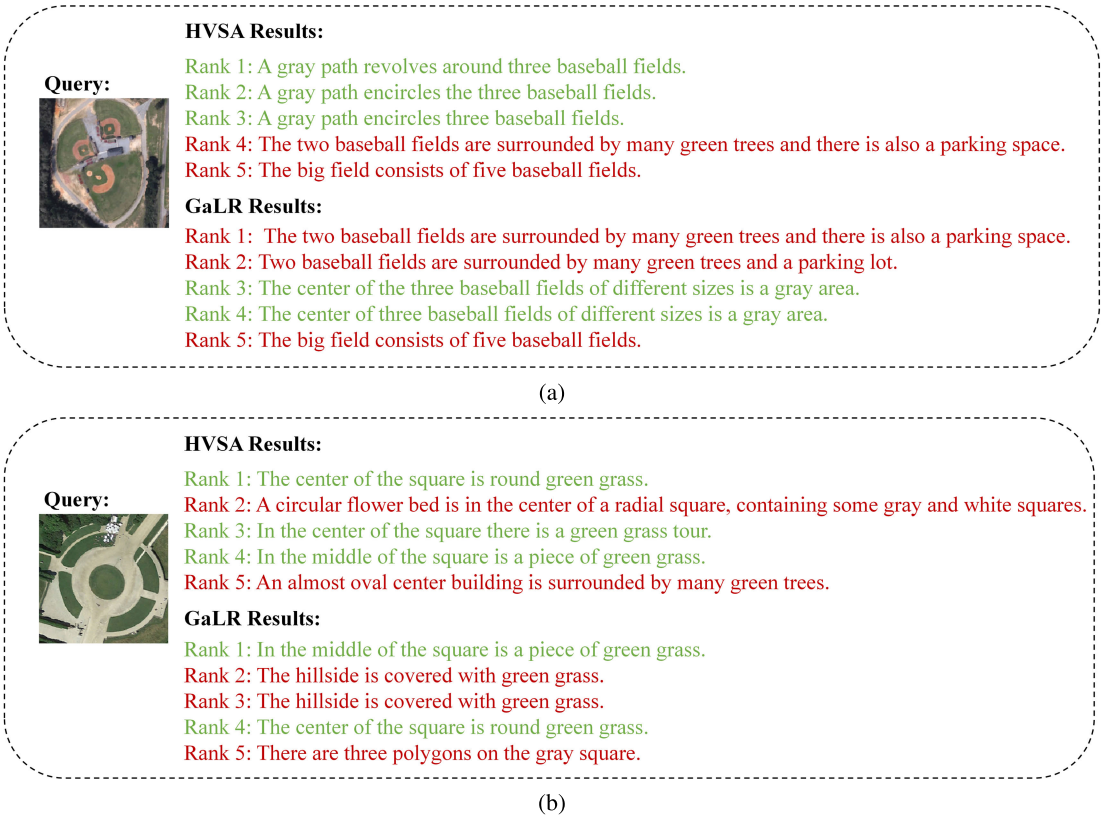


Fig. 12. Qualitative comparison of HVSA and GaLR for image-to-text retrieval on the RSITMD test set. For each image query, we exhibit the top five retrieved captions. Correctly matched captions are highlighted in green, while mistakenly matched captions are highlighted in red.

TABLE V
TIME CONSUMPTION OF DIFFERENCE METHODS

Approach	Evalution time(s) RSICD	RSITMD	Params(M)
VSE++	22.85	5.71	25.79
AMFMN	43.86	10.61	36.70
GaLR	44.25	10.82	46.89
HVSA	28.71	7.33	35.01

performance of our method is verified through qualitative analysis of the retrieval results.

H. Analysis of Time Consumption

We compare the retrieval time consumption of our method with other outstanding methods. The computing platform we use consists of an Intel[1] Xeon[1] Gold 6133 CPU at 2.50 GHz and a single NVIDIA 32G V100 GPU. We choose the evaluation time of the method on the entire test set as our evaluation metric. To test the reliability of the results, we averaged three consecutive experiments. We choose some open source methods that do not precompute the testing data, and the results of the comparison are shown in Table V.

As Table V shows, our proposed method has a slight increase in time consumption compared with VSE++, but our method achieves the current state-of-the-art results. Compared with the classical method AMFMN and GaLR, our concise model structure greatly reduces the time consumed for inference. By comparing the above methods, our proposed method not only has the best performance but also has an acceptable inference time.

[1]Registered trademark.

V. CONCLUSION

In this article, we present an HVSA network via CL. To address the diversified image distribution and high similarity of texts in RSCTIR, we propose an adaptive alignment strategy based on CL. Combining the distributional characteristics of RS cross-modal data and mining the difficulty information of triplet samples make our method more efficient.

By optimizing the uniformity of different semantic features on the unit hypersphere, the generalization of cross-modal feature alignment is improved. Under the restriction of our proposed uniformity loss, feature vectors with different semantic information are distributed as uniformly as possible, which reduces the probability of false matching. To deal with the inherent problem of imbalanced image and text information, we design the KEA mechanism to capture the crucial objects in the RS scene for better matching with the corresponding textual descriptions. KEA is able to obtain a better visual representation by filtering out redundant information and emphasizing valuable entities.

Extensive experiments are conducted on two classical datasets, RSICD and RSITMD, to evaluate the effectiveness of the proposed model. The results demonstrate that our approach

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excels several previous advanced methods while enjoying faster inference. We also perform visualization experiments to verify our model's interpretability, demonstrating its efficiency and generalization through the t-SNE algorithm and attention heatmaps' visualization.

In the future, it might be instructive to explore the RS data features and model the relationships of valuable entities. There are many kinds of entities and complex relationships in RS images, and it is worthwhile to mine the valuable semantic information. In our future work, we hope to combine more fine-grained information for better modeling and reasoning of the relationships among key objects in RS images.

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